12 Rehan is on holiday in the USA.

He has $200 to spend on clothes.

Rehan buys

1 pair of trainers costing $60
3 T-shirts costing $25 each.

He also wants to buy a jacket costing $80

(a) Has Rehan got enough money to buy the jacket?
You must show how you get your answer.

T-shirts: 3 × $25 = $75

Trainers + T-shirts: $60 + $75 = $135

$200 – $135 = $65

$65 < $80, so no; he is $15 short.

3 marks total
Find $135 spent
Find $65 left
Not enough for $80

(3)

The trainers cost $60
The exchange rate is $1 = £0.749

Rehan says,

"The trainers cost less than £40"

Rehan is wrong.

(b) Using a suitable approximation, show working to explain why.

Use 0.7 as a lower estimate of 0.749.

$60 × 0.7 = £42 > £40.

Actual cost exceeds £42, so Rehan is wrong.

2 marks total
Valid lower estimate
Even £42 > £40

(2)

(Total for Question 12 is 5 marks)